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Linkai — Phase 1 Decision Record

This document captures every meaningful decision made during Phase 1 (Ideation & Scoping). It is intentionally narrative rather than a bullet list — the **reasoning** behind each decision matters as much as the decision itself, because Phase 2 (Architecture) will revisit several of these and you need the original justification on hand to push back against scope drift or premature re-litigation.

When in doubt during Phase 2 or implementation: ****re-read this document before changing scope.****

1. The Thesis (Locked)

> Existing branching AI chat tools treat conversational context as either fully inherited or fully isolated — a binary that wastes tokens or loses information. Linkai introduces structured, graph-aware context composition. When a user branches from a highlighted span within a parent response, the child branch receives a hierarchically composed context: the span at full fidelity, the parent at medium fidelity, and progressively older ancestors at decaying fidelity. Beyond a configurable depth threshold, ancestor content is compressed into summaries with retrieval drill-down — a two-tier memory system where summaries serve as the default and full text is retrieved on demand when relevance scoring justifies the cost. The system is evaluated against linear-context and full-inheritance baselines using an LLM-judge framework on a curated multi-turn task set, measuring answer quality per token consumed.

This paragraph is the project's contract with itself. It is the README opener. It is the answer to "tell me about your project" in an interview. Every architectural decision in Phase 2 must serve it.

Why this thesis (and not the original pitch)

The initial pitch was framed around UI features: branching, context toggle, graph visualization. The reframe shifted the project's intellectual center to **context management as the thesis, branching as the mechanism.** This shift matters because:

- The target audience (AI-native companies — Anthropic, OpenAI, Perplexity, Cursor, AI-focused YC startups, and FAANG AI infra teams) is not impressed by chat UIs. They evaluate candidates on their ability to reason about hard, unsolved AI infrastructure problems. Context management is one of those problems.
- The competitive analysis (gitchat, Prompt-Tree, treeGPT, and others surveyed) confirmed that every existing tool in the space has converged on a "each message is a node" data model and a binary inherit/isolate context policy. None of them do span-level branching. None of them have a principled context composition strategy. None of them publish evaluation numbers.
- Reframing the project around context composition turns it from a UI experiment into a systems-and-AI-infrastructure project, which is the resume positioning we want.

Load-bearing terms in the thesis

Each of these is something an interviewer can grab onto. Be ready to defend each one independently.

- **Graph-aware context composition** — the context sent to the model is constructed by walking the conversation graph, not by concatenating a linear history.
- **Hierarchical fidelity decay** — content closer to the branching point is represented at higher fidelity; content further away decays toward summary.
- **Two-tier memory** — summaries are the default representation beyond a depth threshold; full text is retrieved on demand.
- **Retrieval drill-down** — a relevance-scoring step decides when to replace a summary with the original text.
- **Quality per token** — the evaluation metric. Not raw quality, not raw cost. The ratio.

2. Competitive Landscape (Summary)

Full analysis is in the separate competitor notes. The strategic takeaways:

- **Convergence on message-as-node:** Every tool surveyed (gitchat, Prompt-Tree, treeGPT, and HN-surfaced tools) models each user/AI exchange as a node in a canvas or tree. None of them branch *within* a response.
- **No span-level branching:** This is the unfilled gap in the surface-level UX space. The hypothesis is that nobody has filled it because the *context semantics* of sub-response branching are hard, not because the UI is hard.

- **No published evaluation:** None of the surveyed tools publish quality metrics, A/B comparisons, or any rigorous measurement of their context strategies. This is the single largest opportunity for differentiation. Linkai will publish an evaluation report as part of MVP.
- **Prompt-Tree is the strongest existing player.** Its three-layer memory system (user profile, memory bank, folder doc) is genuinely sophisticated. Linkai is not trying to out-feature Prompt-Tree; it is trying to be more *principled* — fewer features, but with a defensible architecture and measured outcomes.

Scope Amendment

Amendment 2026-05-18: SpanOnly strategy added to the evaluation set after discovering that span-anchored branches often involve context-independent questions. Justification: strengthens the evaluation framework with no implementation cost (it's a degenerate case of the existing strategies).